

HANCHU ZHOU

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EDUCATION

University of California, Davis Ph.D. student in Electrical and Computer Engineering	Sep. 2022 - Present Davis, CA
Zhejiang University Bachelor's degree in Control Science (GPA: 3.89)	Sep. 2018 - Jun. 2022 Hangzhou, China

RESEARCH INTERESTS

World Model; Autonomous Driving; Robotics;

SKILLS

Programming:	Python, C, C++, MATLAB, HTML, Markdown, C#, JavaScript
Machine Learning:	JAX, PyTorch, TensorFlow, Scikit-learn, Pandas, OpenCV, Huggingface
Tools:	Open3D, YOLO, Wandb, Matplotlib, Git

RESEARCH PROJECTS

IN-RIL: Interleaved Reinforcement and Imitation Learning Dec. 2024 - May. 2025
arXiv: [IN-RIL](#)

- Proposed a new training approach by using Imitation Learning and Reinforcement Learning in an interleaving way to ensure the learning stability and generalization capability.
- Conducted extensive experiments on various robotics benchmarks and demonstrated the superior performance of our interleaving learning approach.

CarDreamer: World-Model-Based Autonomous Driving Platform Apr 2024 - Jul. 2024
IEEE IoT-J: [CarDreamer](#) → ★220 stars

- Developed the first open-source platform for WM based autonomous driving in response to the lack of an easy-to-use WM based learning platform, with the goal of advancing research in this field.
- Designed various scenarios for training with great flexibility on modality, observability, and intention sharing, enabling users to explore diverse experiments with tailored settings. The trained DreamerV3 models show significant efficiency and safety in each scenario.

World-Model-Based Hierarchical Planning Mar 2024 - Jun. 2024
Under review in JMLR: [HANSOME](#)

- Introduced HANSOME, a WM based hierarchical planning with semantic communication framework, aiming to address the challenges posed by partial observability and non-stationary in autonomous driving.
- Proposed AdaSMO, an entropy-controlled adaptive training method, to balance the training of higher-level and lower-level policies. The extensive experiments show that HANSOME outperforms SOTA WM-based methods in challenging driving tasks and enhances overall traffic safety and efficiency.

RDT-1B Foundation Model Fine-tuning Mar. 2025 - Now
Ongoing research

- Designed a fine-tuning framework that enhances the performance of the VLA foundation model RDT-1B through auxiliary tasks, improving its representation of system dynamics.
- Conducting experiments in the ALOHA simulation environments to boost the downstream task performance of the fine-tuned foundation model.

CommWorld: Communicative World Models for Distributed RL Feb. 2024 - May. 2024

- Proposed CommWorld, the world model with lightweight information sharing, to mitigate the issue of insufficient information in distributed RL. Leveraging world models' generalization capability, CommWorld achieves better performance in the challenging planning experiments.